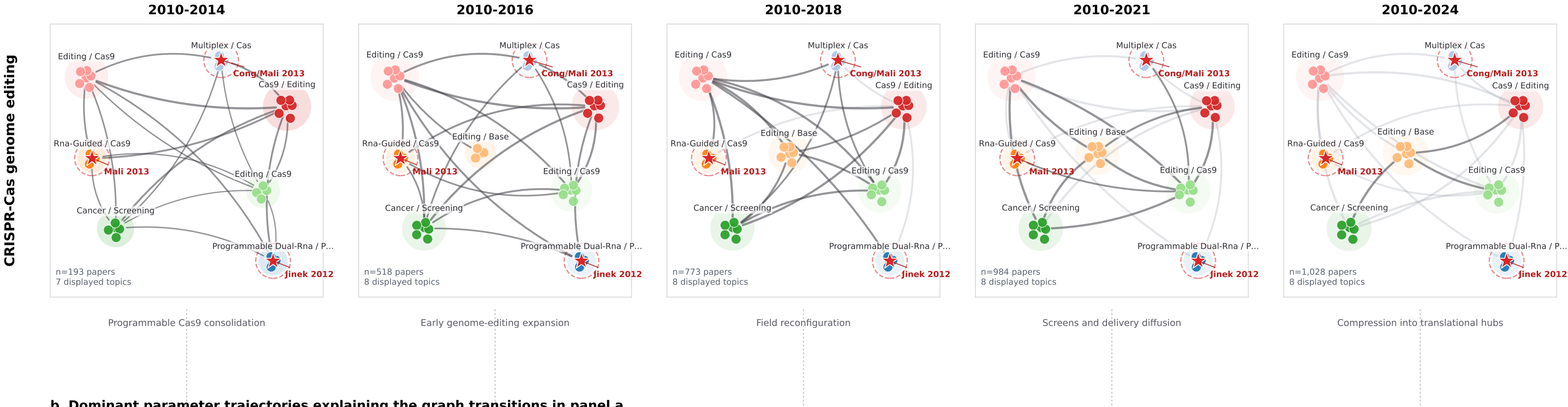


Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

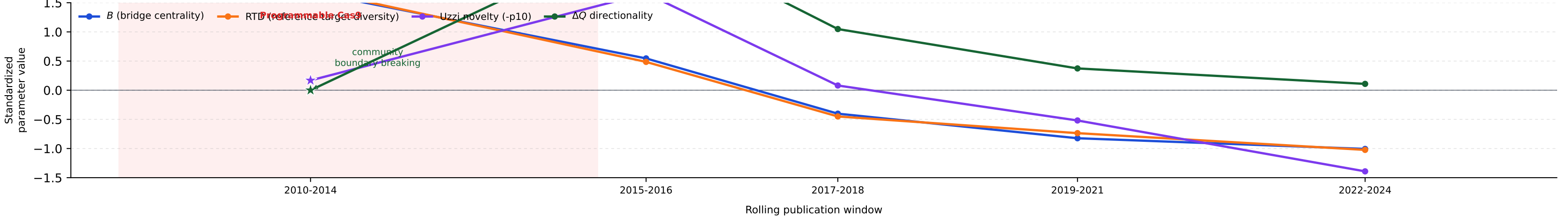
Example domain: CRISPR-Cas genome editing. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots



b Dominant parameter trajectories explaining the graph transitions in panel a



Bridging
B + RTD up
Cross-community paths

Novelty
Uzzi up; DeltaQ < 0
Non-typical recombination

Boundary breaking
DeltaQ < 0
Community borders disrupted

Convergence
B/RTD sustained
DeltaQ near 0